C2W Hybrid Bonding Interconnect Technology for Higher Density and Better Thermal Dissipation of High Bandwidth Memory

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Abstract - Cu/Oxide Hybrid Bonding Interconnect (HBI) technology has been used in the mass production of CMOS Image Sensors and advanced packages for fine pitch heterogeneous interconnection. Furthermore, HBI can be implemented for high density products demanding good heat dissipation due to HBI's structure lacking micro bumps and polymer underfill. Thus, HBI is widely studied in the field of semiconductor assembly. Recently, we have demonstrated HBI technology in chip-to-wafer (C2W) multi stack High Bandwidth Memory (HBM) real devices and obtained promising results. We developed wafer supporting system materials to maintain Cu pad clean and focused on optimizing plasma activation conditions for high bond strength and minimizing change in Cu recess level for stable Cu/Cu joints even in low temperature annealing. Even a single void induced by various particles can cause electrical failures and reliability problems. In C2W, particles can be introduced by various processes and environments prior to bonding, unlike in wafer on wafer (WOW). Hence, an all-in-one stacking equipment with a pre-treatment function was developed to prevent such particles. Through several core technological improvements, we were able to obtain better cumulative yields than anticipated from HBM 8Hi real devices. Although further researches on C2W HBI technology are needed, we have obtained highly reliable HBM through HBI. HBI implemented HBM showed good electrical functions similar to that of conventional HBM, but had a thermal resistance at less than 20% of that of conventional HBM. In addition, we confirmed that HBI can be a powerful solution for 16hi HBM since the stack height can be reduced by more than 15% of conventional HBM. Since HBI has many advantages, we expect it to be widely used in 3D/2.5D packages in the near future.

Keywords – Hybrid bonding, Chip to Wafer Bonding, High Bandwidth Memory, Thermal Resistivity

I. INTRODUCTION

The importance of HBM is rapidly growing alongside the recent development of markets demanding high performance computing, such as Artificial Intelligence, Internet of Things and machine learning. Until now, HBM has been connected through micro solder bumps between chips, but it is required to reduce pitch in order to cope with higher bandwidth. And chip thickness should be thin to make high capacity memory. HBI could be potential candidate for such next generation HBM, as Cu pads and the thin polished passivation layer of HBI can replace the micro solder bumps and polymer underfill, scaling down the bump pitch. Compared to the conventional solder bump structure, HBI can reduce the stacked thickness by 10~15% as shown in Fig.1 or allow for an increase in chip thickness to enable a more stable wafer level processing. However, HBI has yet to overcome problems for adoption to HBM, as HBM requires a high (12+) level of C2W stacking. Unlike wafer to wafer bonding process, chip to wafer HBI should has backside pads for multi-chip interconnection. So HBI should focus on Cu pad management because Cu pad can be easily deteriorated during many thermal processes after front side CMP. Emphasis should also be given to what is trivial at low stacks, but is consequential at high stacks, such as stack voids by particles and cumulative erosion. HBI for HBM is still at the development stage, and may be immature for mass production. But the images and data in this paper are from real HBM devices, not test vehicles, providing concrete evidence that the mass production of HBM using HBI is highly plausible. We hope that this paper could provide insights to diagnose the possibility of HBM in HBI.

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C2W 混合键合互连技术可提高高带宽存储器的密度和更好的散热性

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Abstract - 铜/氧化物混合键合互连 (HBI) 技术已用于大规模生产 CMOS 图像传感器和细间距异构互连的先进封装。此外，由于HBI的结构没有微凸块和聚合物底部填充，因此HBI可以应用于需要良好散热的高密度产品。因此，HBI在半导体组装领域得到了广泛的研究。最近，我们在芯片到晶圆（C2W）多堆栈高带宽存储器（HBM）实际设备中演示了HBI技术，并获得了可喜的结果。 我们开发了晶圆支撑系统材料来保持铜焊盘清洁，并专注于优化等离子体激活条件以实现高键合强度，并最大限度地减少铜凹槽水平的变化，即使在低温退火中也能实现稳定的铜/铜接头。即使是由各种颗粒引起的单个空隙也可能导致电气故障和可靠性问题。 在 C2W 中，与晶圆对晶圆 (WOW) 不同，在键合之前可以通过各种工艺和环境引入颗粒。 因此，开发了一种具有预处理功能的一体式堆垛设备来防止此类颗粒的产生。 通过多项核心技术改进，我们能够从 HBM 8Hi 实际设备中获得比预期更好的累积良率。 虽然C2W HBI技术还需要进一步研究，但我们已经通过HBI获得了高度可靠的HBM。 HBI 实现的 HBM 显示出与传统 HBM 类似的良好电气功能，但热阻不到传统 HBM 的 20%。此外，我们确认 HBI 可以成为 16hi HBM 的强大解决方案，因为堆叠高度可以比传统 HBM 降低 15% 以上。由于HBI具有许多优点，我们预计它在不久的将来将广泛应用于3D/2.5D封装中。

Keywords – Hybrid bonding, Chip to Wafer Bonding, High Bandwidth Memory, Thermal Resistivity

一、简介

随着人工智能、物联网和机器学习等高性能计算市场的最新发展，HBM 的重要性正在迅速增长。到目前为止，HBM一直通过芯片之间的微型焊料凸块进行连接，但需要减小间距以应对更高的带宽。为了制造大容量存储器，芯片厚度应该很薄。 HBI 可能是下一代 HBM 的潜在候选者，因为铜焊盘和 HBI 的薄抛光钝化层可以取代微型焊料凸块和聚合物底部填充，从而缩小凸块间距。 与传统的焊料凸点结构相比，HBI 可以将堆叠厚度减少 10~15%，如图 1 所示，或者允许增加芯片厚度，从而实现更稳定的晶圆级加工。然而，HBI 尚未克服采用 HBM 的问题，因为 HBM 需要高 (12+) 级别的 C2W 堆叠。与晶圆到晶圆键合工艺不同，

芯片到晶圆 HBI 应具有用于多芯片互连的背面焊盘。因此，HBI 应重点关注铜焊盘管理，因为铜焊盘在正面 CMP 后的许多热工艺中很容易劣化。还应强调在低烟囱时微不足道但在高烟囱时却很重要的问题，例如颗粒和累积侵蚀造成的烟囱空隙。 HBM 的 HBI 仍处于开发阶段，对于大规模生产可能还不成熟。 但本文中的图像和数据来自真实的 HBM 设备，而不是测试车辆，

提供了具体证据，证明使用 HBI 大规模生产 HBM 是非常可行的。我们希望本文能够为诊断 HBI 中 HBM 的可能性提供见解。

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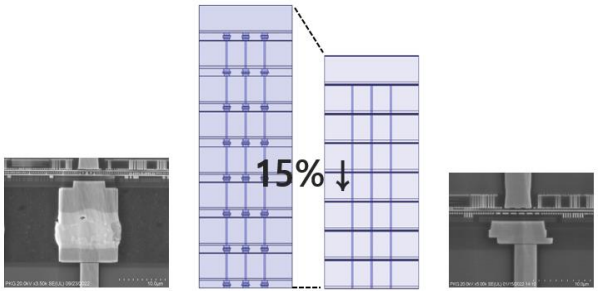


Fig.1 Structure comparison between Solder bump and HBI (in case of same chip thickness of 58um)

II. RESULTS

A. Material Development and Process Opimization for Temporary Bonding Process

For multi-stack chips like HBM, temporary bonding and debonding (TBDB) process is required because bumps or pads must be formed on the front and back side of each layer. There are several methods and materials used for conventional solder bump structure. However, since Cu pads in HBI can be easily oxidized, materials and process conditions need be carefully implemented.

A cure step during temporary bonding process should be required for stable back side processing. Fig.2 shows Cu pad quality after wafer de-bonding process, at different cure conditions. After long annealing time and high temperature, the Cu pad deformed to a protruding topology, which led to deteriorated inter-passivation bond strength. A short annealing time at mid-temperature led to Cu pad oxidation and thus discoloration. As discolored Cu pads form poor Cu/Cu joints, adhesives used to temporarily bond the Cu pad interfaces should be chosen carefully. We established main parameters for optimal Cu pads, such as annealing times, curing temperatures, Q-time between front side CMP and debonding as well as additives for Cu oxidation inhibition.

Materials	A	B	B'	C	C
Cure Conditions	5min@210'C	2hrs@190'C	10min@200'C	10min@200'C	2hrs@190'C
Cu Recess Variance	None	+60nm	None	None	+10nm
Cu Oxide					

Fig. 2 Optical image after de-bonding by TBDB conditions

B. Plasma Treatment Process Optimization for Reliability and Long Q-time at stacking process

Plasma treatment is a main process that makes oxide bonding at room temperature with CMP in HBI. In the early development stage, oxide bonding with low bonding strength could not endure mechanical stress under reliability conditions especially in moisture sensitivity level (MSL) test and highly accelerated temperature and humidity stress test (HAST). So continuous processing with 2 types of gases which increases the OH^- density at the interface was applied without roughness increase of passivation layer. However, edge voids were detected in the stacked chips when the Q-time between the plasma process and the stack process exceeded 4hrs. Thus, plasma treatment parameters were adjusted such that the samples were void and delamination free even at a time window of 24hrs, as shown in Fig.4. Above bonding strength is measured by the Maszara test in wafer to wafer base.

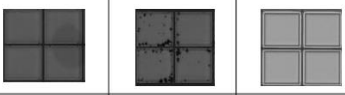
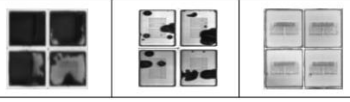
Plasma Conditions		A	B	C
Bond Strength		1.0J	2.5J	2.0J
SAM	End of Line @ TV			
	After Reliability @ Real device			

Fig. 4 SAT Results of End of line and reliability test by plasma conditions

C. Cu Recess Management for Cu/Cu Joint at Low Temp. Annealing

Annealing conditions for HBM in HBI should be limited to below 200℃ to prevent the over-growth of base wafer solder intermetallic compounds, and refresh degradation of DRAM. Data from Scanning Probe Microscopes showed that Cu pads with 10um diameter expanded vertically by approximately 10nm at 200℃, as shown in Fig.5. So Cu recess should be carefully controlled to prevent joint failure.

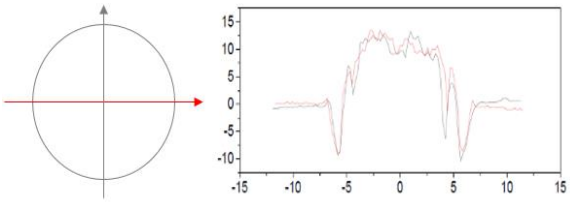


Fig.5 Measurement of Cu Pop-up at 200'C

So Cu recess should be carefully controlled to prevent joint failure. Joints failed at Cu recess level of -4nm during /after the CMP process. Measurements of Cu recess for each step between front side CMP and stacking confirmed that Cu recess grew rapidly during the mega-sonic cleaning process, a process performed to remove particles. Fig.6 also shows a positive correlation between dishing and Cu recess. CO_2 in De-ionized water, used to prevent electro-static discharge contributed to additional recess formation and Fig.6 also shows that the larger dishing is, the more recesses is occurred.

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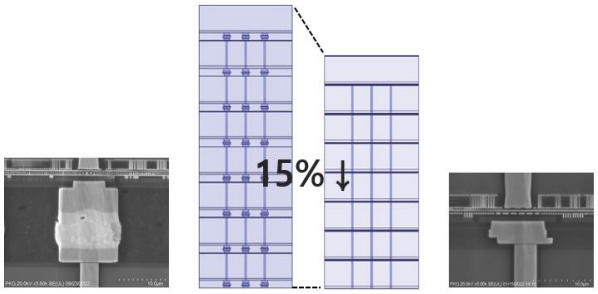

图1 焊料凸块与HBI结构对比（芯片厚度相同58um情况下）

二.结果

A. Material Development and Process Opimization for Temporary Bonding Process

对于像 HBM 这样的多层芯片，需要临时键合和脱键合 (TBDB) 工艺，因为必须在每层的正面和背面形成凸块或焊盘。传统焊料凸块结构有多种方法和材料。然而，由于 HBI 中的铜焊盘很容易被氧化，因此需要仔细实施材料和工艺条件。

为了实现稳定的背面处理，临时粘合过程中需要进行固化步骤。图 2 显示了在不同固化条件下晶圆脱粘工艺后的铜焊盘质量。经过长时间的退火和高温后，铜焊盘变形为凸出拓扑，导致钝化间结合强度下降。中温退火时间短会导致铜焊盘氧化，从而变色。由于变色的铜垫会形成不良的铜/铜接头，因此应仔细选择用于临时粘合铜垫界面的粘合剂。我们建立了最佳铜焊盘的主要参数，例如退火时间、固化温度、正面 CMP 和脱粘之间的 Q 时间以及铜氧化抑制添加剂。

Materials	A	B	B'	C	C
Cure Conditions	5min@210℃	2hrs@190℃	10min@200℃	10min@200℃	2hrs@190℃
Cu Recess Variance	None	+60nm	None	None	+10nm
Cu Oxide					

图2 TBDB条件下脱粘后的光学图像

B. Plasma Treatment Process Optimization for Reliability and Long Q-time at stacking process

等离子处理是在HBI中通过CMP在室温下进行氧化物键合的主要工艺。 在早期开发阶段，键合强度较低的氧化物键合无法承受可靠性条件下的机械应力，特别是在潮湿敏感性等级（MSL）测试和高加速温湿度应力测试（HAST）中。 所以采用两种气体进行连续处理，增加了界面处的OH^-密度，而没有增加钝化层的粗糙度。 然而，当等离子体工艺和堆叠工艺之间的Q时间超过4小时时，在堆叠芯片中检测到边缘空隙。 因此，调整等离子体处理参数，使得即使在 24 小时的时间窗口，样品也没有空隙且无分层，如图 4 所示。 上述键合强度是通过晶圆到晶圆基底的 Maszara 测试来测量的。

Plasma Conditions		A	B	C
Bond Strength		1.0J	2.5J	2.0J
SAM	End of Line @ TV			
	After Reliability @ Real device			

图 4 等离子条件下的 SAT 下线结果和可靠性测试

C. Cu Recess Management for Cu/Cu Joint at Low Temp. Annealing

HBI 中 HBM 的退火条件应限制在 200℃ 以下，以防止基底晶圆焊料金属间化合物过度生长以及 DRAM 的刷新退化。扫描探针显微镜数据显示，直径为10um的铜焊盘在200℃,时垂直膨胀约10nm，如图5所示。 因此，应仔细控制铜凹陷，以防止接头失效。

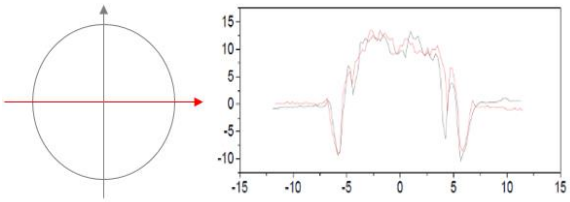

图5 200’C时Cu弹出的测量

因此，应仔细控制铜凹陷，以防止接头失效。在 CMP 工艺期间/之后，接头在 -4nm 的铜凹进水平处失效。对正面 CMP 和堆叠之间每个步骤的铜凹槽测量证实，铜凹槽在兆声波清洗工艺（用于去除颗粒的工艺）中快速生长。图 6 还显示了凹陷和 Cu 凹陷之间的正相关关系。去离子水中的CO_2，用于防止静电放电导致额外凹陷的形成，图6还显示凹陷越大，出现的凹陷越多。

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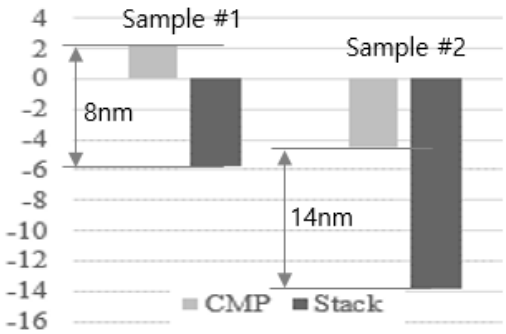

Fig.6 Cu Recess results according to process step

So de-ionized water with reduced CO_2 concentration was used, and the cleaning time was shortened to minimize the Cu recess. After optimization of the cleaning and CMP conditions, Cu/Cu joints did not fail even at low temperature annealing. It is expected to be additionally required of special cleaning method not affecting Cu recess but removing particles easily.

D. Minimization of Stack Voids

Even a single void can cause electrical failure and reliability problem. In the HBI stack process, void can be generated by various reasons, including outgassing of passivation film during the annealing process, and uneven flatness during stack process. Additional processes such as pre-bake used in the c-mos image sensor field to prevent outgassing are difficult to implement, and uneven flatness is exacerbated, due to the multi-layer stacked structure of the HBM package. As an alternative, we have developed a low-outgassing passivation film, redesigned the Cu pad, and optimized CMP conditions such as the slurry and process parameters.

However, the biggest cause of stack voids is air trap formed during the stack process, originating from different kinds of particles and sequential bonding failure form the center. In the early stage of development, a suitable convex tool that could initiate the center of chip could easily remove the void trap. However, in real chip with large warpage and uneven flatness, having perfect flatness, stack void could not be removed by the hard type convex tool. After experiments to optimize main factors such as tool design and material properties, the air trap induced stack voids could be removed as shown in Fig.7. In addition, adjustments to bond parameters including force, time and speed were required for the real chips to be stack void free.

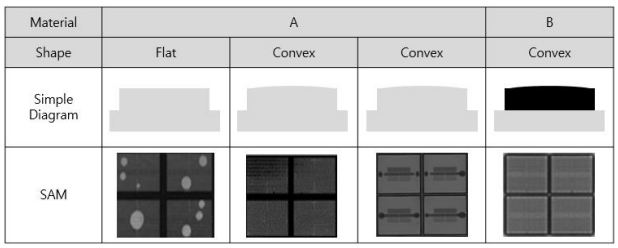

Fig.7 Air trap according to bond tool

Unlike chips with bumps or die attach films, HBM with HBI has no gaps, and particles lead to voids. Scanning Acoustic Tomography shows that thin and soft particles may not cause voids, but on signal pads, particles results in voids that cause yield losses, as shown in Fig.8.

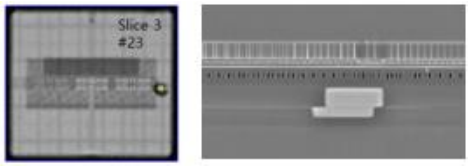

Fig.8 SAM with failure map vs. failure analysis results

In case of HBI process, many kinds of foreign materials were occurred by dicing process including stealth dicing and expand-cut. Also, environmental particles could be originated by storage and even transportations as shown in Fig.9.

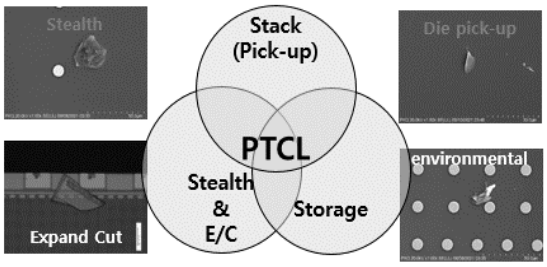

Fig. 9 Major foreign materials

Dicing process – The front side of wafer should be contacted on chuck table during stealth dicing and silicon debris cannot be avoidable due to expand-cut process. Most of such foreign materials originated by dicing process could be easily removed by applying protection layer during dicing process and high pressure cleaning after dicing process. However, when the residue of the protection layer remains after processes, it leads to yield loss regardless of residue size. Therefore, we are still working on improving the solubility of protection layer and finding new cleaning methods.

Stacking process – Chip to wafer stacking process involves die pick-up, chip transfer and bonding, which requires repeated transport of individual dies, and a long processing time unlike wafer to wafer bonding process. This increases the risk of contamination from continuous movement and prolonged exposure time. In particular, the pick-up area is much more contaminated than other areas, as shown in Fig.10.

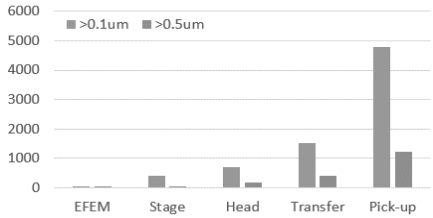

Fig.10 the number of particles generated by area in C2W Bonder

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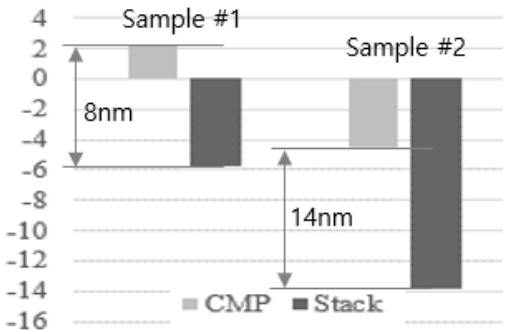

图 6 根据工艺步骤的 Cu Recess 结果

因此使用降低CO_2浓度的去离子水，并缩短清洗时间以最大限度地减少Cu凹陷。优化清洁和 CMP 条件后，即使在低温退火下，Cu/Cu 接头也不会失效。 预计还需要不影响Cu凹部但容易去除颗粒的特殊清洁方法。

D. Minimization of Stack Voids

即使是单个空隙也会导致电气故障和可靠性问题。 在HBI堆叠工艺中，空洞的产生可能有多种原因，包括退火过程中钝化膜的脱气以及堆叠过程中平整度不均匀等。由于 HBM 封装的多层堆叠结构，c-mos 图像传感器领域中用于防止排气的预烘烤等附加工艺很难实现，并且加剧了平坦度不均匀的情况。 作为替代方案，我们开发了低释气钝化膜，重新设计了铜垫，并优化了 CMP 条件，例如浆料和工艺参数。

然而，堆叠空隙的最大原因是堆叠过程中形成的气穴，其源自不同种类的颗粒以及从中心开始的连续键合失败。在开发的早期阶段，合适的凸形工具可以启动切屑中心，可以轻松去除空洞陷阱。 然而，在实际芯片中，翘曲较大，平整度不均匀，即使具有完美的平整度，硬质凸面工具也无法去除堆叠空隙。经过对工具设计和材料特性等主要因素进行优化的实验后，气穴引起的叠层空隙可以被消除，如图 7 所示。 此外，需要调整粘合参数，包括力、时间和速度，才能使真正的芯片堆叠无空隙。

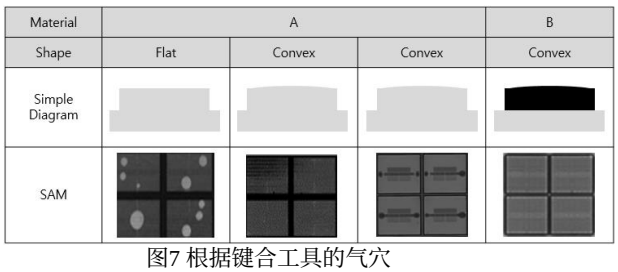

图7 根据键合工具的气穴

与带有凸块或芯片贴膜的芯片不同，带有 HBI 的 HBM 没有间隙，颗粒会导致空隙。扫描声学断层扫描显示，细而软的颗粒可能不会导致空洞，但在信号焊盘上，颗粒会导致空洞，从而导致良率损失，如图 8 所示。

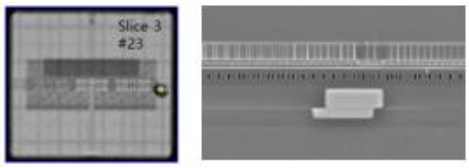

图8 带有失效图的SAM与失效分析结果

在HBI工艺中，包括隐形切割和扩切在内的切割工艺会产生多种异物。此外，环境颗粒可能源自储存甚至运输，如图 9 所示。

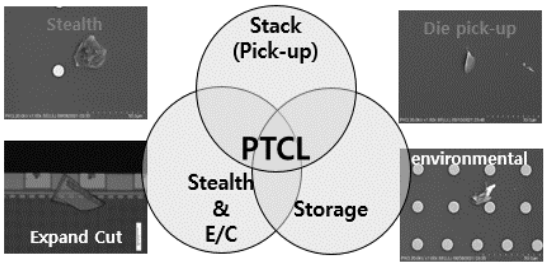

图9 主要异物

切割工艺 – 在隐形切割过程中，晶圆的正面应接触卡盘台，并且由于扩展切割工艺，硅碎片是不可避免的。大多数由切割过程产生的异物可以通过在切割过程中施加保护层以及切割过程后的高压清洗来容易地去除。然而，当工艺后保留保护层的残留物时，无论残留物的大小如何，都会导致良率损失。因此，我们仍在致力于提高保护层的溶解度，寻找新的清洁方法。

堆叠工艺——芯片到晶圆堆叠工艺涉及芯片拾取、芯片转移和键合，与晶圆到晶圆键合工艺不同，这需要重复运输单个芯片，且处理时间较长。 这增加了连续移动和长时间暴露时间造成污染的风险。 特别是，接送区域比其他区域污染严重得多，如图10所示。

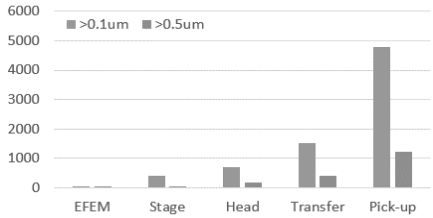

图10 C2W Bonder中按面积产生的颗粒数

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Thus, a particle suction at the pick-up area, and a particle blowing at the bonding area could be able to lower yield losses induced by particles. However, remounting process, which can fundamentally remove particles generated form diced wafers was much more efficient for stack void reduction. but it cannot but agree that innovative elimination method should be developed.

All-in-one stacking equipment – It takes a few hours to manufacture 1 wafer of 8 high stack. However, as the clean room classification for assembly sites is much more lenient compared to fabrication sites, stacking poses a risk of contamination when the wafers are processed for a long time. We have set-up and used all-in-one stacking equipment to handle pre-treatment process and stacking process altogether. This minimizes particle induced stack voids originating from transportation and storage, as the cleaning process is applied immediately before HBI bonding. In addition, this reduces the cycle time, and eases lot management.

E. Reliability Test Results and Thermal and Electrical Performance of HBI

HBM must withstand complex follow-up processes to be a part of SIP, and thus is inevitably subjected to more mechanical stress than conventional packages. To validate the structural robustness of HBI, a pre-condition test that surpasses moisture sensitivity level 2, and a unbiased-HAST test has been conducted to verify robustness of HBI. The results showed no additional delamination even for voids of 1mm in diameter in HBM cubes with oxide bonding strength of $2J/m^2$.

However, in case of proxy package for system in package, HBI was subjected to more stress due to the greater difference in Coefficient Thermal Expansion (CTE) at the side mold area, than compared to packages with underfills, as shown in Fig.11. Low CTE Epoxy Molding Compound (EMC) was able to release stresses at the triple point between EMC and chips, so it was found that delamination did not occur.

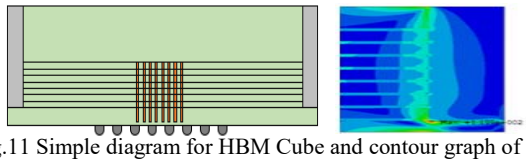

Fig.11 Simple diagram for HBM Cube and contour graph of stress

The HBI structure reduces the interconnection length by more than 100um compared to that of the 8 high micro bump structure, and reduces parasitic interconnection resistance by 19%, inductance by 35% and capacitance by 15%. Thus, the HBI structure can reduce the channel loss from of LC/RC delay in the interconnection. However, the structure of HBM2E exhibits no difference in electrical performance when compared to existing micro bump structures, in 176 evaluation items shown in the table 1.

Table. 1 Electrical performance comparison between HBI and MR-MUF

	Items	Number of Evaluation Items	HBI	MR-MUF
CHDV	IDD0~8	13	22.3W	22.1W
	tCK	4	4.17Gbps	4.17Gbps
	MRS	3	Spec in	Spec in
	Async	21	Spec in	Spec in
	PDN	17	Spec in	Spec in
IFDV	Input Shmoo	10	Spec in	Spec in
	Output Shmoo	3	Spec in	Spec in
FUDV	HBM1	17	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen1	5	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen2	16	> 4.0Gbps	> 4.0Gbps
	HBM2E Func.	63	> 4.0Gbps	> 4.0Gbps
Total		176		

Fig.12 shows that HBM cube with HBI has lower thermal resistance than the current HBM with micro bumps and molded underfills. Even if under-fill materials with high thermal conductivity (~0.7W/℃) and bumps of high density are used, HBM with HBI still outperforms current HBM. Fig.12 shows data that when the bump densities are the same, the HBI structure has a 22.8% lower thermal resistance

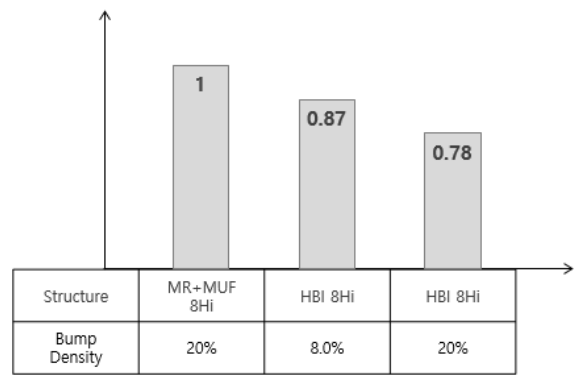

Fig.12 Thermal Resistance Comparison according to each structure (Molded Underfill vs. HBI)

III. CONCLUSIONS

In this paper, we demonstrated 8Hi-HBI packages successfully using the latest real HBM. To obtain robust Cu/Cu joints at low annealing temperatures, special glues for the TBDB process were developed. And CO_2 content were controlled to manage the level of Cu recess between CMP and stacking process aside from precise parameters control of CMP. Also, plasma conditions were optimized for stronger passivation bonds and longer Q-time. As a single void can cause electrical failure and reliability issues, many corrective actions including stacking parameter and particle reduction have been applied to lower stack void. And Low CTE EMC was essential along with strong bonding strength of passivation layers for robust HBI package Finally, HBI has been proven to have no problems in electrical characteristics in real devices manufactured at low annealing temperatures. HBM with HBI

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因此，拾取区域处的颗粒吸力以及粘合区域处的颗粒吹气能够降低由颗粒引起的良率损失。 然而，重新安装工艺可以从根本上去除切割晶圆产生的颗粒，对于减少堆叠空隙更加有效。但它不能不同意应该开发创新的消除方法。

一体式堆叠设备——制造 1 片 8 层堆叠的晶圆需要几个小时。然而，由于组装场所的洁净室分类比制造场所宽松得多，因此当晶圆长时间加工时，堆叠会带来污染风险。 我们设置并使用一体式堆垛设备来完成预处理过程和堆垛过程。 由于在 HBI 粘合之前立即进行清洁工艺，因此可以最大限度地减少因运输和存储而产生的颗粒引起的堆叠空隙。此外，这还缩短了周期时间，并简化了批次管理。

E. Reliability Test Results and Thermal and Electrical Performance of HBI

HBM 必须承受复杂的后续工艺才能成为 SIP 的一部分，因此不可避免地会比传统封装承受更多的机械应力。 为了验证 HBI 的结构稳健性，进行了超过湿度敏感度 2 级的前提条件测试和无偏 HAST 测试来验证 HBI 的稳健性。结果表明，即使对于氧化物结合强度为 2J/m^2 的 HBM 立方体中直径为 1mm 的空隙，也不会出现额外的分层。

然而，在封装系统的代理封装的情况下，与带有底部填充的封装相比，由于侧模具区域的热膨胀系数 (CTE) 差异更大，HBI 承受更大的应力，如图 11 所示。 低CTE环氧模塑料(EMC)能够释放EMC和芯片之间的三相点处的应力，因此发现没有发生分层。

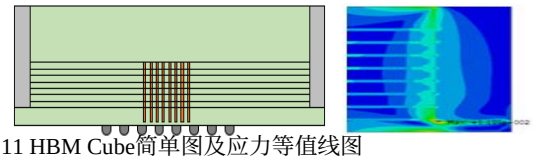

图11 HBM Cube简单图及应力等值线图

HBI结构与8高微凸块结构相比，互连长度减少100um以上，寄生互连电阻降低19%，电感降低35%，电容降低15%。 因此，HBI结构可以减少互连中LC/RC延迟造成的通道损耗。 然而，在表1所示的176个评估项目中，HBM2E的结构与现有的微凸块结构相比，在电性能上没有差异。

桌子。 1 HBI与MR-MUF的电气性能比较

	Items	Number of Evaluation Items	HBI	MR-MUF
CHDV	IDD0~8	13	22.3W	22.1W
	tCK	4	4.17Gbps	4.17Gbps
	MRS	3	Spec in	Spec in
	Async	21	Spec in	Spec in
	PDN	17	Spec in	Spec in
IFDV	Input Shmoo	10	Spec in	Spec in
	Output Shmoo	3	Spec in	Spec in
FUDV	HBM1	17	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen1	5	> 4.0Gbps	> 4.0Gbps
	HBM2 Gen2	16	> 4.0Gbps	> 4.0Gbps
	HBM2E Func.	63	> 4.0Gbps	> 4.0Gbps
Total		176		

图 12 显示，具有 HBI 的 HBM 立方体比当前具有微凸块和模制底部填充的 HBM 具有更低的热阻。 即使使用高导热率（~0.7W/℃）的底部填充材料和高密度凸块，具有HBI的HBM仍然优于当前的HBM。图12数据显示，当凸块密度相同时，HBI结构的热阻降低了22.8%

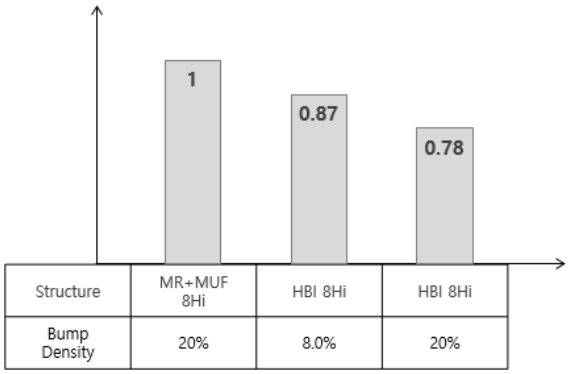

图 12 各结构的热阻比较（模制底部填充与 HBI）

三．结论

在本文中，我们使用最新的真实 HBM 成功演示了 8Hi-HBI 软件包。为了在低退火温度下获得坚固的铜/铜接头，开发了用于 TBDB 工艺的特殊胶水。 除了精确控制 CMP 参数外，还控制 CO_2 含量以管理 CMP 和堆叠工艺之间的 Cu 凹进水平。 此外，还优化了等离子体条件，以获得更强的钝化键和更长的 Q 时间。 由于单个空隙可能导致电气故障和可靠性问题，因此已采用包括堆叠参数和颗粒减少在内的许多纠正措施来降低堆叠空隙。低 CTE EMC 以及钝化层的强大结合强度对于坚固的 HBI 封装至关重要。最后，事实证明，在低退火温度下制造的实际器件中，HBI 的电气特性不存在任何问题。 HBM 与 HBI

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has a lower thermal resistance than that stacked with micro bumps and underfills.

Although much of the stack voids were removed, a yield loss of over 10% was unavoidable due to foreign materials. We expect new problems to arise in 12 or 16 layer-high stacked HBM, but as many integrated device manufacturing companies, equipment suppliers and material manufacturers interested in multi-stack HBI, we believe that multi-stack HBM with HBI will be mass-produced in the near future.

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与微凸块和底部填充堆叠相比，具有更低的热阻。

尽管大部分堆叠空隙被消除，但由于异物的影响，超过 10% 的良率损失是不可避免的。 我们预计12层或16层堆叠HBM会出现新的问题，但由于许多集成器件制造公司、设备供应商和材料制造商对多堆叠HBI感兴趣，我们相信具有HBI的多堆叠HBM将在不久的将来实现量产。

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